

NANYANG TECHNOLOGICAL UNIVERSITY
SEMESTER 2 EXAMINATION 2022-2023
CE4172 – Internet of Things: Tiny Machine Learning

Apr/May 2023

Time Allowed: 1 hour

INSTRUCTIONS

1. This paper contains 5 questions and comprises 2 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. Questions do NOT carry equal marks.

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1. (a) State four benefits of TinyML based applications. (10 marks)
 - (b) Explain the main feature of inference operations that makes them suitable for TinyML deployment. (15 marks)
 2. Describe and explain the steps used to deploy a trained TensorFlow model on microcontroller such as the Arduino Nano 33 sense board. (15 marks)
 3. Briefly describe the underlying operating process of an audio keyword detection application in TinyML during inference. (25 marks)
 4. State and explain two methods that can be used to reduce the size of the model for faster inference. (10 marks)

5. (a) State the advantages of Depthwise Separable Convolution such as those used in MobileNet architecture.
(10 marks)
- (b) Why is Depthwise Separable Convolution not suitable for TinyML implementations?
(15 marks)

END OF PAPER

CE4172 INTERNET OF THINGS: TINY MACHINE LEARNING

Please read the following instructions carefully:

- 1. Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.